Muhamed Rizwan Mehaboob

LinkedIn | Gmail | GitHub

Education

2023 – 2027 B.Tech. in Data Science and Engineering – IIT Palakkad (Pursuing)
2023 Class 12th CBSE – Chavara Public School, Pala (96.2%)

Technical Experience

Software Engineering Intern - HSBC Technology India Jun 2026 – Jul 2026

- Contributed to the Global Mortgage Solutions platform, working across a large-scale multi-region application ecosystem.
- Worked on engineering initiatives involving application maintenance, internal tooling and improving development workflows.
- Built and enhanced internal solutions to support application analysis and make software changes more efficient and reliable.

Student TA - Administration Section, IIT Palakkad Feb 2026 – Ongoing

- Developing a guest house booking portal with role-based workflows for booking, approval and document management.
- Building a visitor management system with role-based access, QR-code verification and digital entry passes.
- Working closely with the Administration Section to streamline and digitize existing campus processes.

Technical Projects

IIT Palakkad Campus Hub – PostgreSQL, React, Express, Airflow, dbt, Docker Oct 2025 – Aug 2026

- Built a full-stack campus hub platform serving IIT Palakkad students, with an Airflow + dbt pipeline loading a dimensional star schema and 4 automated data-quality checks, containerised via Docker Compose.

Arcus – Academic Retrieval Workspace – Next.js, tRPC, PostgreSQL, pgvector, Prisma, Inngest Mar 2026 – Aug 2026

- Built a document RAG platform with hybrid retrieval (pgvector HNSW + Postgres full-text search) fused via Reciprocal Rank Fusion, with an idempotent ingestion pipeline and benchmark evaluation using Recall@k, MRR, and nDCG.

Multi-Target Drug Molecule Generation – Python, PyTorch, RDKit, MXNet, ProtTrans Apr 2026 – Ongoing

- Developing a pipeline for multi-target, multi-property drug generation using protein embeddings and molecular representations.

Multilingual Disaster Message Classification – Python, Transformers, PyTorch Aug 2025 – Nov 2025

- Built a multilingual transformer-based NLP model to classify disaster messages and prioritize urgent emergency requests.

Guest House Booking Portal – Next.js, React, TypeScript, Supabase, PostgreSQL, Tailwind Aug 2026 – Ongoing

- Engineering a guest house booking portal with multi-tier approvals, dynamic forms and an interactive availability console.

Image Super-Resolution using Deep Neural Networks – Pytorch, Python, Deep Learning Jan 2026 – May 2026

- Developed deep learning models including **SRCNN**, **SRGAN**, and custom CNN-based models to convert low-resolution images into high-resolution images while preserving visual details.

Positions of Responsibility

Class Representative, Data Science (Aug 2023 – Dec 2023, Aug 2024 – Dec 2024)
Hospitality Head, Petrichor 2026, IIT Palakkad (May 2025 – May 2026)
Core Team Member, Industry Academia Conclave, IIT Palakkad (Sep 2024, Sep 2025, Sep 2026)

Technical Skills

Languages : Python, C++, C, PostgreSQL, MySQL, MongoDB
Frameworks/Libraries : Scikit-learn, Pandas, NumPy, PyTorch, Transformers
Tools : Git, GitHub, Linux, VS Code, Jupyter

Relevant Coursework

Core CS : Database Management Systems, Computer Systems, C-Programming, Data Structures and Algorithms
Core DS : Intro to AI, Machine Learning, Deep Learning, Big Data Lab, Natural Language Processing, Data Analytics

Achievements

- Represented IIT Palakkad at Inter IIT Tech Meet 14.0 for the Adobe Problem Statement.
- Selected as a Semifinalist in Flipkart GRiD 8.0 in the AI Engineering Track, competing among participants nationwide.